# Yan Jiahao

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## Education

**Renmin University of China, Gaoling School of Artificial Intelligence, Artificial Intelligence Major** **Beijing, China**

2024.9

- GPA： Overall GPA: 3.92/4.00 (Rank: 4/80). Since entering the Gaoling School of Artificial Intelligence, GPA: **4.00/4.00 (Rank: 1/80;** the second-ranked student has a GPA of 3.90).
- The paper **ManuSearch** was accepted to **EMNLP 2025 and has received 15 citations** to date. The paper **MemMamba** is currently under review focusing on rethinking memory patterns in state space models for long-sequence modeling.The paper **HSI** introduces a stealthy backdoor attack targeting the 3D memory architecture of LLM agents, proposed in March 2026.
- Academic Performance:： Mathematical Analysis 4.0 (95), C Programming 4.0 (93), Artificial Intelligence and Python Programming 4.0 (95), Probability and Mathematical Statistics 4.0 (96), Data Structures and Algorithms 4.0 (90), Computer Systems Fundamentals 4.0 (95), Introduction to Artificial Intelligence (Honors) 4.0 (92), Mathematical Foundations of Artificial Intelligence (Honors) 4.0 (94), Principles of Economics 4.0 (93),
- Awarded the "Pioneer Medal" of Renmin University of China

## Research Experience

- Joined the research group of Professor Zhao Xin （Wayne Xin Zhao） [google scholar] at the Gaoling School of Artificial Intelligence in the second semester of the freshman year. 2025.3

  **ManuSearch**: Democratizing Deep Search in Large Language Models with a Transparent and Open Multi-Agent Framework

  Closed-source LLM search systems monopolize complex reasoning, while open-source lacks transparent, modular alternatives. We propose ManuSearch, a multi-agent framework decoupling search into **Planner, Searcher, and Reader,** and introduce the **ORION benchmark** for long-tail entity reasoning.

  **Output**:This paper was accepted to **EMNLP 2025** and has received **15 citation**s to date.

- In collaboration with Shanghai AI Lab | the second author 2025.11

  **MemMamba**: Rethinking Memory Patterns in State Space Model

  **Transformers'** quadratic complexity hinders truly long contexts, while **SSMs** (e.g., Mamba) achieve linear time but suffer rapid long-range memory decay. We propose the "horizontal-vertical memory fidelity" metric to quantify information degradation, and introduce MemMamba: a state summarization module (Note Block) plus lightweight cross-layer and cross-token attention, preserving salient information dynamically without sacrificing linear complexity, thus **alleviating long-range forgetting**.

- Initial Thoughts | the first Author 2026.3

  **HSI**: A Stealthy Backdoor Attack Targeting the **3D Memory Architecture** of Large Language Model Agents

  3D memory architecture evades existing attacks; defenses have blind spots. HSI decomposes malicious policy into linked fragments, exploits clustering and summarization to infer full rules, distills them into consensus, and activates via general questions to manipulate decisions.

## Competition & Activity Experience

2025 China Undergraduate Mathematical Contest in Modeling – Beijing First Prize (Team Leader) 2025.8

## Extracurricular Activities

1. Minister, Student Union of Gaoling School of Artificial Intelligence 2025.9
2. Head of Shanxi 01 Group, "RUC Ambassadors" Project, Renmin University of China 2025.3
3. Participated in a research and practice activity in Guizhou, visiting the National Big Data Comprehensive Pilot Zone Exhibition Center, and engaged in in-depth discussions with local government officials from Guizhou as a student representative. 2026.1
4. Visited the headquarters of Yushu Technology in Hangzhou to understand the current state and practical demands of the AI industry. 2026.3